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Second year report: Identifying higher order graph structures using motifs

Introduction

Graph theory explores the inter-relationships between abstract discrete objects, and how we can organise them to describe real world phenomena, from brain networks to financial structures [1] [2]. My PhD explores the various ways in which we can use induced subgraphs, called motifs, to calculate motif adjacency matrices based on graphs which emphasise and highlight different structural topological features within graphs. My research thus far has explored how we can use motifs in 3 distinct high-level directions:

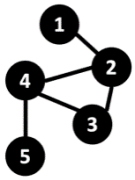
- To identify community structures within graphs.
- To identify bipartite community structures within graphs.
- To explore centrality within graphs.

Background

Each discrete object in a graph is called a vertex, v_i, with the set of all vertices $V = \{1,2,\dots,v_n\}$, where $|V| = n$. The connections between these vertices are called edges, $E \subseteq V \times V$. Using this list of edges we can construct an adjacency matrix, A, wherein

$$a_{i,j} = \begin{cases} 1 & if\ (v_i, v_j) \in E, \\ 0 & if\ (v_i, v_j) \notin E. \end{cases}$$

This relationship is intuitive with a visual example, and so one is shown in Figure 1.



$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix},$$

Figure 1) shows the adjacency matrix of a graph and its visual representation.

Using this information, we will be able to define a graph object $G = (V, E)$, where V is the vertex list and E is the edge list. Graph objects may have repeated edges between two vertices giving $a_{i,j} > 1$ and may have reflexive properties, allowing edges to exist between a vertex and itself with $(v_i, v_i) \in E$. We can construct directed edges as $(v_i, v_j) \in E$ and $(v_j, v_i) \notin E$, however this will make our adjacency matrix non-symmetric. For my PhD I have constrained the set of graphs I will consider working with to be those that are non-reflexive and do not contain multiple edges between vertices.

Each vertex of a graph has a degree, which is simply the sum of edges it belongs to

$$d_i^{(out)} = \sum_{j=1}^{n} a_{i,j} \text{ or } d_i^{(in)} = \sum_{j=1}^{n} a_{j,i}.$$

If the graph is directed, these two measures of degree will not be equal, and instead $d_i^{(out)}$ would represent the out-degree (the degree of connections pointing out of a given node) and $d_i^{(in)}$ represents the in-degree (degree of connections to a given node). If $\forall(i,j) \in E \exists(j,i) \in E$ and $|E| > 0$ then the graph is undirected, and we can refer to the degree as d_i.

The degree of the vertices of a graph can be used to construct a degree matrix, $D = \text{diag}(d_1, \dots, d_n)$, which we can use to construct a new representation of a graph called a graph Laplacian, $L = D - A$, or the normalized Laplacian, $L = I - D^{\frac{1}{2}}AD^{\frac{1}{2}}$. A graph Laplacian is useful for many things, including detecting community structures across a graph, and will be explored further later. However, the graph Laplacian is undefined in the case of directed graphs which limits the methods that can be used to discover communities on directed graphs.

A motif is a small graph, $G_M = (V_M, E_M)$, that may occur repeatedly in G. We can count the occurrences of the graph G_M in G, by finding subgraphs of G, G_S which are isomorphic to G_M. Hence for $S \subseteq V$, if $G_S(S, E_S), (E_S = (S \times S) \cap E)$ is isomorphic to G_M, then we say that S is an occurrence of G_M in G. We denote by $OCC_{G_M}(G)$ the set containing all occurrences of G_M in G. Using this set we can construct our motif adjacency matrix (MAM):

$$MAM_{i,j} = |S \in OCC_{G_M}(G)|$$

Again, this is a concept best explained graphically with the example in Figure 2, in which we can identify two occurrences of the motif G_M shown in Figure 2.b) in the graph G shown in figure 2.a); one occurrence in the vertices {1,2,4} and another in {5,4,3}. As these two sets are both occurrences of the motif G_M, their corresponding nodes in the MAM are fully connected as shown in Figure 3.c). Note that {2,4,3} in 1.a) is not an occurrence of G_M because of the edge from vertex 2 to vertex 3 in G.

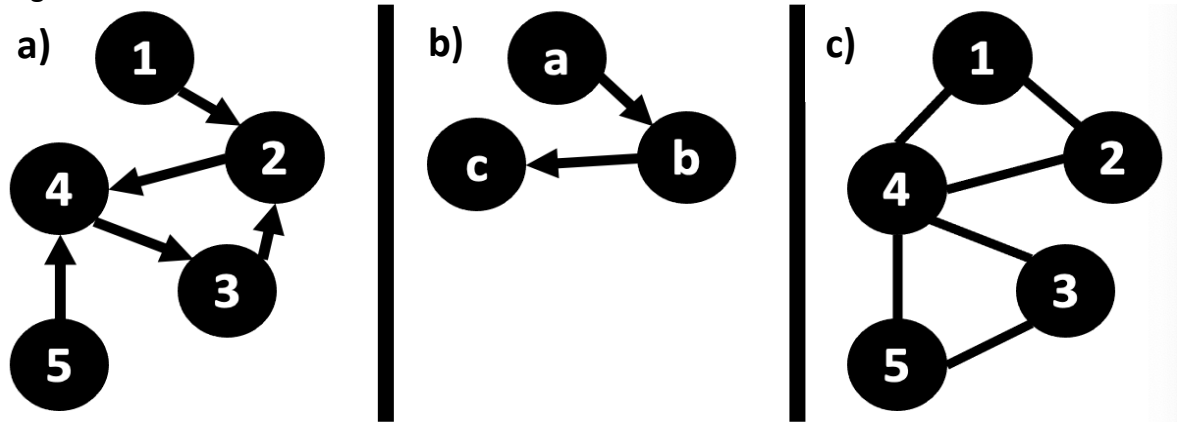


Figure 2a) Directed graph G. 2b) Motif G_M, 2c) Motif adjacency graph given by the occurrences of the motif G_M in G

Anchoring a motif is an extension to the previously discussed strategy for limiting the number of edges added to our MAM when we find occurrences of a motif within a graph. We consider $\alpha \subseteq V_M$ as the anchor set of a motif and after we find an instance of G_M we add only edges in the MAM over the vertices in A instead of over all V_S. Consider the previous example wherein we identified occurrences of the motif in Figure 2b) in the graph in Figure 2a) to produce the MAM in Figure 2c). If we use this example with motif in Figure 2b) subject to the anchors $\{a, c\}$, then after we've found our occurrences {1,2,4} we would add edge (1,4) to our MAM and then with the occurrence {5,4,3}, edge (5,3) would be added to

the MAM. To bring this altogether we will look at how we can use motif adjacency matrices to produce partitions which have community structure.

Community detection

Community detection is the process of partitioning of a graph into sets $\{s_1, s_2, \dots, s_n\}$ such that the number of within community (intracommunity) edges is maximised whilst the number of between community (intercommunity) edges is minimised. When detecting just two communities within a graph, s_1 or s_2, this process takes the form of an optimisation problem over the labels $\{-1,1\}$ for each $\{i,j\} \in E$ in a graph $G = (V, E)$ such that:

$$\min_{v \in \{-1,1\}^k} f(v) = \sum_{(i,j) \in E} (v_i - v_j)^2$$

where, if an edge occurs such that the vertex v_i is in set $\{1\}$ and v_j is in set $\{-1\}$, then a cost of 4 is added. Thus a method that attempts to produce a partition that minimises this cost, minimises the number of intercommunity edges across the graph G. Fiedler et al showed on a relaxed version of the problem, allowing $v \in \mathbb{R}$, that assigning labels according to the sign of the elements of the eigenvector corresponding to the second smallest eigenvalue of the Laplacian of G produces a partition that minimises the aforementioned cost [3]. A similar and often used cost is conductance [4], derived from the probability that a random walker on the graph will walk from a vertex in set 1 to a vertex in set 2. This cost is very similar to the previous one, but weights the sum based on the number of vertices that belong to the smallest partition, forcing the size of the two partitions to be as even as possible whilst minimising the number of intercommunity edges;

$$\phi(s_1) = \frac{\sum_{i \in s_1, j \in s_2} a_{ij}}{\min\{a(s_1), a(s_2)\}}, a(s_n) = \sum_{i \in s_n} \sum_{j \in V} a_{ij}.$$

Some examples of conductance on an example graph for three partitions are given in Figure 3.

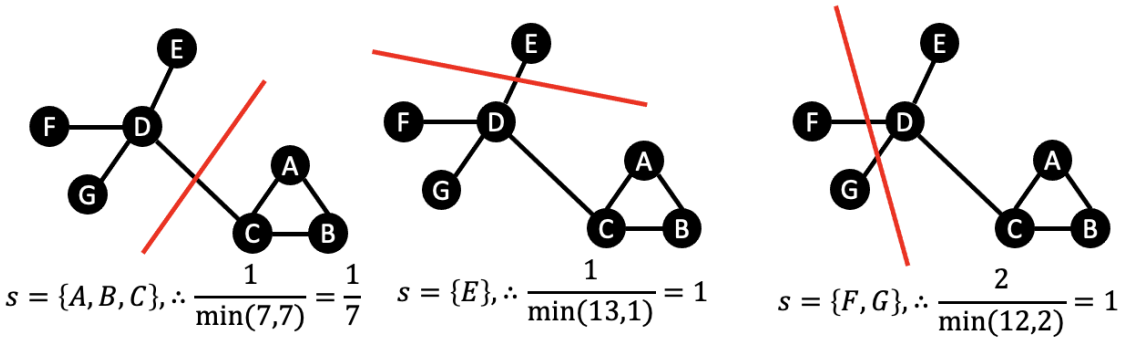


Figure 3 Conductance calculations for 3 different partitions on the same graph.

Most importantly Benson showed that, for the case of three-node motifs, the partition that minimised the conductance across the motif adjacency matrix could also minimise the conductance across the graph [4]. Whereas previously, directed graphs had been quite difficult to partition into communities (due to reconciling the directions of the edges and having non-symmetrical Laplacians), using motifs provides an avenue to allow the use of undirected methods on directed graphs. Thus, by taking the partition that minimises the

conductance across the motif adjacency matrix for a directed motif we can produce a partition of a directed graph. However, in the rest of this section we will consider examples of only undirected graphs.

In some instances, we have shown that you can improve the partitioning of the MAM by Fiedler vector by taking a summation of multiple motif adjacency matrices for different motifs, to create a representation that includes many different structural topologies. We can emphasise commonly occurring structural phenomena by thresholding the motif adjacency matrix based on the median weighted value, mitigating noise introduced by non-frequently occurring instances of the subgraph. To test this out we'll explore how we can calculate partitions using the Fiedler vector on the Laplacian of the MAMs of triangle (T238) and wedge (T78) motifs compared to calculating the Fiedler vector on the graph Laplacian of the Zachary karate club dataset [5].

Method	No Motif	Motif T78	Motif T238	Motif T238 + 0.1*T78
Overlap with ground truth	94.12%	70.59%	50%	97.06%
% of NNZ	100%	440%	86%	530%

Table 1 Shows the overlaps between the partitions generated using various methods compared to the ground truth of the Zachary karate club network. % of NNZ shows the number of non-zero elements relative to the adjacency matrix as a percentage

The idea we have presented here is simply that triangles have a denser structure and thus are more sparsely distributed within the graph but represent key anchors for our community structure. On the other hand, wedges occur relatively frequently, ensuring that the MAM includes every vertex of the adjacency matrix. Furthermore, wedges are heavily represented in stars, providing central structures for our partitioning to be based on, ideally overlapping with those that also include occurrences of triangles.

It is not shown here, but in general this philosophy expands to directed graphs using directed motifs. Even more central is the previously explored concept of anchors in directed motifs, which allow you to selectively pick out key topological features to emphasise in the partitioning. We note that whilst there are only two unique undirected three-node motifs there are 11 directed three-node motifs. Further the number of unique motifs gets larger as the size of motif increases. However, directed motifs will not be considered in the following section as bipartite community structure is defined strictly for undirected graphs.

Bipartite community detection

A bipartite network is a special type of network which can be decomposed into two disjoint sets such that there do not exist any edges between nodes within the same set. A near-bipartite graph is one in which there exist a "few" intracommunity edges between the two sets of vertices (the meaning of a "few" is contingent on the application). This is an anti-community structure, wherein we are attempting to maximise the number of intra-community edges whilst minimising the number of inter-community edges. Given a random partition of a graph into two sets, we can assess how close to bipartite a partition is by using modularity, which is commonly used and is defined as follows [6]:

$$Mod(A,\{s_1,s_2\}) = \frac{1}{2m}\sum_{ij}\left(A_{ij} - \frac{k_i k_j}{2m}\right)\frac{s_i s_j + 1}{2}$$

where k_i is the expected degree of node v_i and m is the total sum of all edges in A. Modularity defined in this form is upper bounded at 1 if the partition over the graph has a community structure and -0.5 if the partition of the graph has a bipartite structure. To use MAMs to produce bipartite community structures we need to select motifs that have a bipartite structure with the correct anchoring, for example consider the example in Figure 4.

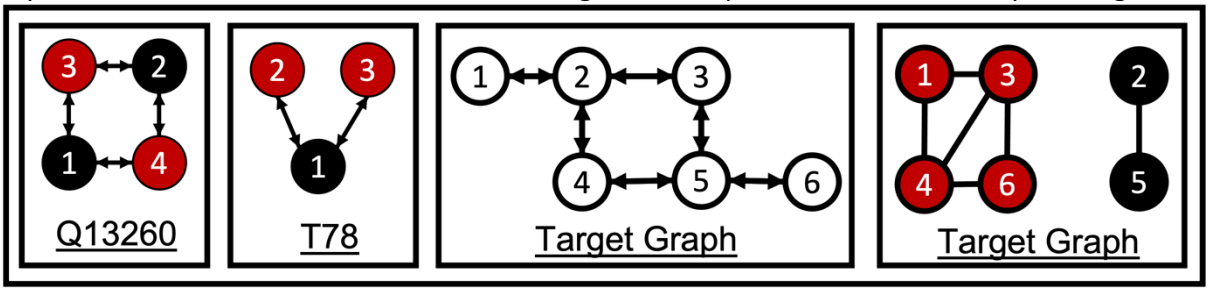
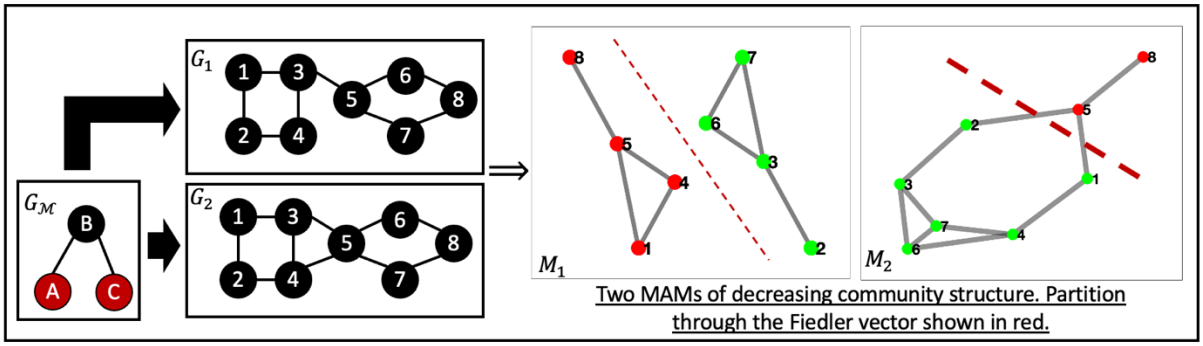


Figure 4 shows two motifs, Q13260 and T78, that have bipartite structures (set 1 shown as red, set 2 as black), a target graph bidirectional edges and finally the motif adjacency matrix showing each independent community as each of the bipartite sets in the original graph.

The Motifs Q13260 and T78 are the only examples of a motif with a bipartite structure in undirected motifs of up to four nodes, and we will use them to generate our MAMs on some real-world data to produce bipartite community structure. In practice, many real-world graphs are not bipartite but are instead near-bipartite, resulting in a graph with some edges between nodes within the same community, resulting in a MAM with intercommunity edges. Depending on the number of intercommunity edges in the MAM this can produce single clustered component, or a MAM with multiple communities. Graphs that are near bipartite but have some triangles present will introduce two key behaviours:

1. Constructive edges – Some non-bipartite edges will introduce occurences of the motif that by their nature are not bipartite, creating intercommunity edges with the MAM which will skew the partitioning towards community detection.
2. Deconstructive edges – Some edges remove critically important occurences of the bipartite motif. Normally this occurs when adding an edge across a quadrangle.

In general, graphs that have many deconstructive edges will be very difficult to produce near-bipartite partitions for using this method whereas graphs with constructive edges add noise that can result in some error but have the general bipartite structure conserved. Figure 5 gives a visual example of both of these phenomena in action.



Two MAMs of decreasing community structure. Partition through the Fiedler vector shown in red.

Figure 5 shows the bipartite motif $G_{\mathcal{M}}$ applied to two very similar graphs (G_1, G_2) with corresponding partitioned MAMs (M_1, M_2). G_2 has an instance of a constructive edge $(4,5)$ resulting in a misclassification of two nodes.

Within Figure 4) we see that when the occurrences of motif $G_{\mathcal{M}}$ are calculated over the graph G_1 we produce a MAM with two disjoint communities whose labels map back to the graph G_1 to produce perfectly bipartite partition. Whereas when the motif $G_{\mathcal{M}}$ is calculated over G_2, the MAM is a single clustered component. The new edge in G_1 acts as both a deconstructive and constructive edge. It removed the occurrence {4,3,5} that was previously in G_1as well as adding the new occurrences {2,4,5}, {4,5,6} and {4,5,7}. This results in the MAM M_2 which is a single clustered components with misclassification of the nodes 1 and 4. Whereas the labels assigned from M_1 produce a partition with modularity -0.5 on G_1, the labels assigned from M_2 onto G_2 produce a partition with modularity -0.18. If, for completeness, we assigned the labels from M_1 to the graph G_2, a partition with modularity -0.405 is produced. This example highlights the impact a single edge can have on the quality of bipartite partition generated from MAMs.

We can avoid some of this behaviour by selectively removing edges that participate in triangles, as these subgraphs are those resulting in most of the intercommunity edges within our MAM. However, by doing so we will produce a MAM with many disconnected components whose label will need to be determined in post. Testing this strategy on 32 datasets we find that we can produce bipartite partitions with close to the same modularity as previous methods [7]. One previous method we will compare to we'll call the A method. This approach determines the clustering of nodes into bipartite sets by assigning labels according to the signs within the eigenvector corresponding to the smallest eigenvalue of the adjacency matrix of the graph. A short summary is provided below the table.

Dataset	MAM method	A method	Dataset	MAM method	A method
Canton	-0.476	-0.476	"PIN_Malaria"	-0.255	-0.250
Caviar	-0.286	-0.276	"Revere"	-0.047	-0.047
CEleg	-0.181	-0.192	"Roget"	-0.200	-0.213
Centrality	-0.216	-0.223	"Scotch"	-0.389	-0.395
ColoSpg	-0.425	-0.451	"SouthFL"	-0.183	-0.183
Corporate	-0.077	-0.079	"Stony"	-0.474	-0.474
Dolphin	-0.224	-0.256	"TM1"	-0.352	-0.401
Drug	-0.172	-0.180	"TM2"	-0.414	-0.460
Dublin	-0.122	-0.123	"TM3"	-0.303	-0.329
El_verde	-0.314	-0.314	"Tortoise"	-0.188	-0.210
Electronic1	-0.299	-0.357	"Trans_EColi"	-0.424	-0.437
Electronic2	-0.333	-0.365	"Trans_Yeast"	-0.461	-0.437
Electronic3	-0.332	-0.341	"US_Air"	-0.163	-0.166
Enron	-0.149	-0.176	"Voles"	-0.188	-0.207
Enz_g87	-0.213	-0.267	"VTK"	-0.325	-0.326
Enz_g355	-0.323	-0.348	"Wildbird"	-0.099	-0.102

Table 2 The modularity of partitions produced using the MAM method after removing triangles using the motif T78_12 + 0.1*Q13260_13 with post-partition correction compared to the partition produced by taking the eigenvector corresponding to the smallest eigenvalue of the adjacency matrix with post-partition correction.

Across these 32 datasets, the MAM method produces partitions with lower modularity in 3 datasets and performs close to the A method in all other cases. Hence this has shown that with the correct selection of motifs, blended together, it is possible to create a MAM whose partition produces a near-bipartite structure on the graph which is competitive with traditional methods and in one instance produces a near-bipartite structure that is even lower modularity than could be previously identified using gold-standard methods [7]. The third and final stream of my research has been investigating to what extent we can use motifs to explore centrality nodes/structures within directed graphs.

Centrality within MAMs

Centrality in the context of graph theory provides strategies to evaluate which vertices within a graph are key to its structural connectivity. Whilst in practice there are many methods for assessing centrality, we will constrain our study to two which emphasize different structural properties of undirected graphs: eigenvector centrality and closeness centrality. Eigenvector centrality is defined from the eigenvector corresponding to the largest eigenvalue of the adjacency matrix.

$$EV_v = \frac{1}{\lambda}\sum_{u\in G} a_{v,u}x_u,$$

where EV_v represents the centrality score for the vth vertex of the graph, and x is the eigenvector corresponding to the largest eigenvalue of A, λ. By contrast the closeness centrality score is based on the reciprocal of the distance between a given vertex v and every other vertex within the graph.

$$CN_v = \frac{N-1}{\sum_u d(u,v)},$$

where $d(u,v)$ denotes the distance from vertex u to v and $N = |V|$. Closeness centrality emphasises nodes that are central to the graph geodesics. Between eigenvector and closeness centrality this gives a good measure of both the nodes central to the span of the network which in the case of MAMs are nodes those that participate in many occurrences of the motif of choice.

Page rank is a method for determining the centrality of vertices within a directed graph based on the number of times a random walker would transition to or from those vertices. Taking each random walker to have an equal probability of starting on each node, the initial centrality score is given by $PR^{(0)} = [\frac{1}{N}, \dots, \frac{1}{N}]^T$ and at each iteration, this score is updated by the transition matrix $\tau_{i,j} = \frac{1}{a_{i,j}} \forall (i,j) \in E$. Hence the page rank orderings can be determined by

$$PR^{(k)} = \tau^k PR^{(0)},$$

where k is some sufficiently large number to ensure the steady state (or close enough to it) is achieved. For each node i in G, its centrality score is given by $PR_i^{(k)}$. More information on page rank can be found in [8].

From the community detection section, we saw that the MAM of some motifs can be significantly less dense than the adjacency matrix of a graph. It is however unknown as to how much of the centrality information of the adjacency matrix is conserved in the MAM and as to whether the MAM can provide a low-density representation of a graph that conserves this centrality information. To test how the centrality scores of motif adjacency matrices of varying directed motifs overlaps with the centrality scores of the adjacency matrix we will compare the overlap between the top 10 elements of the page rank of the adjacency matrix to both the eigenvector centrality rankings and closeness rankings for each motif's MAM (results shown in Table 3).

Motif	Comp_sci	History	Business	Motif	Comp_sci	History	Business
T6	70 | 50	80 | 80	50|40	T78	100 | 50	80 | 60	20|20
T12	40 | 70	40 | 40	0|0	T98	30 | 20	70 | 30	30|40
T14	100 | 90	100 | 90	70|70	T102	70 | 50	60 | 40	30|20
T36	20 | 20	10 | 10	10|10	T108	50 | 50	40 | 40	20|20
T38	40 | 50	80 | 80	10|20	T110	100 | 90	90 | 90	40|50
T46	90 | 100	90 | 90	90|70	T238	100|100	100|100	80|70
T74	30 | 40	50 | 20	0|0	-	-	-	-

Table 3 Percentage overlap between EV|CN for different motifs for various academia-based reputation networks compared to the page rank of the adjacency matrix of that academia network.

Hence from the above the MAMs corresponding to motif T46, T14, T110 and T238 all conserve the majority of the centrality information regarding the top 10 nodes within all three reputation-based networks. Furthermore, all three MAMs have significantly fewer non-zero (nnz) elements relative to the adjacency matrix of the respective graphs.

	Comp_sci	History	Business
%NNZ of T14 cut	80.81%	82.07%	94.91%
%NNZ of T46 cut	62.31%	78.34%	95.60%
%NNZ of T110 cut	25.86%	29.12%	61.43%
%NNZ of T238 cut	7.32%	10.50%	26.86%

Table 4 Non-zero elements remaining after using the MAM to mask various academia-based reputation networks.

Thus, we have shown here that motifs can be used to produce masks that conserve a user defined amount of the centrality information within a graph, whilst cutting a large portion of the number of non-zero elements of the adjacency matrix. Sparsification can be used as a type of community detection; by removing all the edges to and between nodes which are not important we can produce a clustering/community of core nodes which conserves most of the centrality information within the network. Furthermore, if we can reliably recreate the centrality results for much larger graphs, then this method could be used in situations where computing the eigenvector centrality or closeness centrality is not feasible. More research is certainly needed in both areas over the next year of my PhD.

Conclusions and future work

Overall, in this report we have explored how to use motifs to produce motif adjacency matrices for any graph, how we can use MAMs for partitioning of graphs into communities and bipartite structures, and how we can use motifs to identify central nodes within a network subject to a topological constraint. We have shown that MAMs of dense motifs tend to be sparser than the adjacency matrix of a graph, but that MAMs also have the potential to produce very dense representations of graph structure (especially when using the wedge motif T78). We have shown that there is a strong need to select the correct motif for the graph at hand to produce a partitioning with a high level of agreement with the ground truth and that there is a need to automate this process as there are many motifs (13 3-node, 199 4-node motifs) and exhaustive testing is computationally costly. In the future we look to round off some of this work and compile the results into a paper or two and explore how we can use accurately to predict which motif and combinations of motifs we should use to produce optimal clustering.

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